

In  $\triangle HRW$ ,  $m\angle RHW = 32^\circ$ ,  $m\angle HWR = 17^\circ$ , and  $WR = 25$ . Determine the measurement of each part of  $\triangle HRW$ . Show your work below the table. If necessary, round to the nearest tenth.

|    | Part            | Measurement |
|----|-----------------|-------------|
| 1. | $m\angle WRH$   | 131°        |
| 2. | $\overline{HR}$ | 13.8        |
| 3. | $\overline{HW}$ | 35.6        |

1)  $180 - 32 - 17 = 131$

2)  $\frac{\sin 32}{25} = \frac{\sin 17}{HR}$ ;  $25 \sin 17 = HR(\sin 32)$ ;  $\frac{25 \sin 17}{\sin 32} = HR$ ;  $13.8 = HR$

3)  $\frac{\sin 32}{25} = \frac{\sin 131}{HW}$ ;  $25 \sin 131 = HW(\sin 32)$ ;  $\frac{25 \sin 131}{\sin 32} = HW$ ;  $HW = 35.6$

In  $\triangle BTX$ ,  $m\angle TXB = 26^\circ$ ,  $TB = 8$  and  $XT = 11.2$ . Determine the measurement of each part of  $\triangle BTX$ . Show your work below the table. If necessary, round to the nearest tenth.

|    | Part            | Measurement |
|----|-----------------|-------------|
| 4. | $m\angle XBT$   | 37.9°       |
| 5. | $m\angle XTB$   | 116.1°      |
| 6. | $\overline{XB}$ | 16.4        |

4)  $\frac{\sin 26}{8} = \frac{\sin \theta}{11.2}$ ;  $11.2 \sin 26 = 8 \sin \theta$ ;  $\sin^{-1} 0.6137 = \theta$ ;  $\theta = 37.9$

5)  $180 - 26 - 37.9 = 116.1$

6)  $\frac{\sin 26}{8} = \frac{\sin 116.1}{XB}$ ;  $8 \sin 116.1 = XB \sin 26$ ;  $XB = 16.4$

Shevrin and Tracey are both hiking in the Ocala National Forest. They are currently 12 miles apart. They plan to meet up at their campsite.

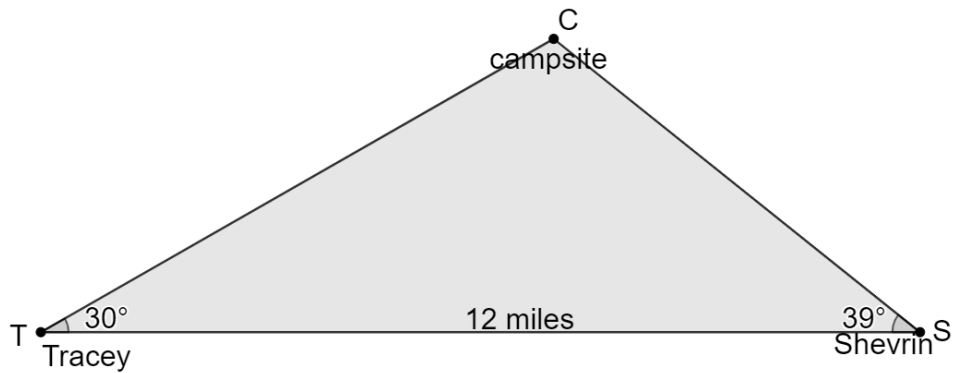
7. How far, to the nearest hundredth of a mile, does Tracey hike?

$$\frac{\sin 39}{x} = \frac{\sin 111}{12}$$

$$x \sin 111 = 12 \sin 39$$

$$x = \frac{12 \sin 39}{\sin 111}$$

$$x = 8.09 \text{ miles}$$



8. How far, to the nearest hundredth of a mile, does Shevrin hike?

$$\frac{\sin 30}{x} = \frac{\sin 111}{12}$$

$$x \sin 111 = 12 \sin 30$$

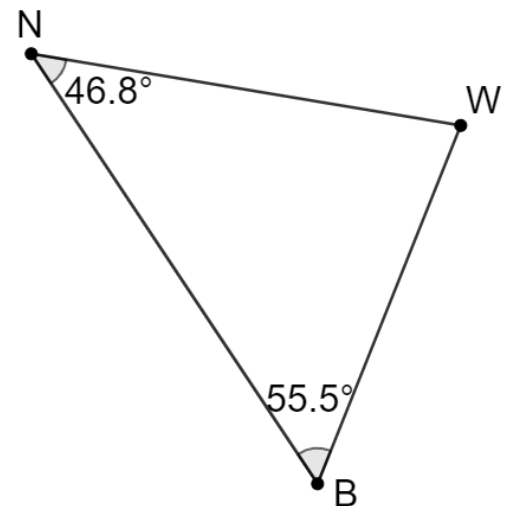
$$x = \frac{12 \sin 30}{\sin 111}$$

$$x = 6.43 \text{ miles}$$

Alejandro flies a drone in a competitive obstacle course. The obstacle course and some of its measurements are shown. The distance between point  $N$  and point  $B$  is 7.2 kilometers.

9. Write an equation that can be used to determine the distance, in kilometers, between point  $N$  and point  $W$ .

$$\frac{\sin 77.7}{7.2} = \frac{\sin 55.5}{x}$$



10. Determine the perimeter, in kilometers, of the obstacle course.

$$x \sin 77.7 = 7.2 \sin 55.5; x = \frac{7.2 \sin 55.5}{\sin 77.7}; x = 6.07 \text{ (NW)}$$

$$\frac{\sin 77.7}{7.2} = \frac{\sin 46.8}{x}; x \sin 77.7 = 7.2 \sin 46.8; x = \frac{7.2 \sin 46.8}{\sin 77.7}; x = 5.27 \text{ (WB)}$$

$$7.2 + 6.07 + 5.27 = 18.54 \text{ km}$$